ESM_262_HW3

Assignment 3

Part 1.

```
# Create sequence of days
day <- 1:365

# Generate realistic daily PM2.5 values (urban air quality)
# Average around 18 µg/m³ with some variability
pm25 <- rnorm(365, mean = 18, sd = 9)

# Prevent negative values
pm25 <- pmax(pm25, 0)

# Create season factor based on day number
season <- ifelse(day <= 90, "winter",
          ifelse(day <= 180, "spring",
          ifelse(day <= 270, "summer", "fall")))

season <- as.factor(season)

# Create empty risk column
risk <- NA

# Combine into data frame
air_quality <- data.frame(
  day = day,
  pm25 = pm25,
  season = season,
  risk = risk
)

# View first few rows
head(air_quality)
```

```
  day      pm25 season risk
1   1 20.750988 winter   NA
2   2  5.850838 winter   NA
```

```
3   3  1.608060 winter   NA
4   4 16.991363 winter   NA
5   5 17.362149 winter   NA
6   6 14.992595 winter   NA
```

Part 2.

```r
#initializing
current_unhealthy_streak <- 0
max_unhealthy_streak <- 0

#risk column
for(i in 1:nrow(air_quality)) {
  air_quality$risk[i] <- ifelse(
    air_quality$pm25[i] < 12, "good",
    ifelse(air_quality$pm25[i] <= 35.4, "moderate", "unhealthy"
  )
 #consequtive unhealthy days
if(air_quality$risk[i] == "unhealthy") {
  current_unhealthy_streak <- current_unhealthy_streak + 1
  if(current_unhealthy_streak > max_unhealthy_streak) {
    max_unhealthy_streak <- current_unhealthy_streak
  }
} else {
  current_unhealthy_streak <- 0
}
}


head(air_quality)
```

```
  day      pm25 season     risk
1   1 20.750988 winter moderate
2   2  5.850838 winter     good
3   3  1.608060 winter     good
4   4 16.991363 winter moderate
5   5 17.362149 winter moderate
6   6 14.992595 winter moderate
```

```r
max_unhealthy_streak
```

```
[1] 1
```

```
air_quality
```

```
    day       pm25 season     risk
1     1 20.7509876 winter  moderate
2     2  5.8508384 winter      good
3     3  1.6080604 winter      good
4     4 16.9913626 winter  moderate
5     5 17.3621488 winter  moderate
6     6 14.9925951 winter  moderate
7     7 31.2799983 winter  moderate
8     8 19.9849841 winter  moderate
9     9 16.7301135 winter  moderate
10   10  9.8318984 winter      good
11   11 12.4602307 winter  moderate
12   12  9.6326474 winter      good
13   13 23.5807609 winter  moderate
14   14 11.0335970 winter      good
15   15 15.7636689 winter  moderate
16   16 14.6204187 winter  moderate
17   17 17.8386809 winter  moderate
18   18 11.0882844 winter      good
19   19 24.8173909 winter  moderate
20   20 17.2556859 winter  moderate
21   21  4.5795384 winter      good
22   22  1.8412520 winter      good
23   23 27.1337368 winter  moderate
24   24 13.9518219 winter  moderate
25   25 31.4436590 winter  moderate
26   26 12.7610118 winter  moderate
27   27 28.6014957 winter  moderate
28   28 21.5563657 winter  moderate
29   29 13.4591274 winter  moderate
30   30 30.9843779 winter  moderate
31   31 24.8903219 winter  moderate
32   32 12.5719617 winter  moderate
33   33 18.1791327 winter  moderate
34   34 18.4505809 winter  moderate
35   35  0.4162726 winter      good
36   36 25.1528900 winter  moderate
37   37 13.7886931 winter  moderate
38   38 14.9134981 winter  moderate
39   39 18.8073980 winter  moderate
40   40 15.6795103 winter  moderate
41   41 18.8953118 winter  moderate
```

```
42   42 16.9238406 winter  moderate
43   43 23.8981722 winter  moderate
44   44 41.4511144 winter unhealthy
45   45 13.9048453 winter  moderate
46   46 16.3623147 winter  moderate
47   47 28.4391094 winter  moderate
48   48 21.1345142 winter  moderate
49   49 29.2791743 winter  moderate
50   50 17.8356859 winter  moderate
51   51 18.1969731 winter  moderate
52   52 20.3350793 winter  moderate
53   53 15.0586013 winter  moderate
54   54 20.2129508 winter  moderate
55   55 16.8910217 winter  moderate
56   56 19.2304619 winter  moderate
57   57 19.7204113 winter  moderate
58   58 12.2520976 winter  moderate
59   59 18.1383924 winter  moderate
60   60 25.5416541 winter  moderate
61   61 13.3677160 winter  moderate
62   62 13.1582075 winter  moderate
63   63 10.5773538 winter      good
64   64 19.7495439 winter  moderate
65   65  9.8357938 winter      good
66   66 21.6515445 winter  moderate
67   67 24.6630681 winter  moderate
68   68 24.9625268 winter  moderate
69   69 15.3461001 winter  moderate
70   70 11.8653153 winter      good
71   71 31.2089353 winter  moderate
72   72  5.9441208 winter      good
73   73 12.2687297 winter  moderate
74   74 33.6323768 winter  moderate
75   75 14.3521071 winter  moderate
76   76 20.5812292 winter  moderate
77   77 17.9825638 winter  moderate
78   78 28.0745410 winter  moderate
79   79 11.0400914 winter      good
80   80 24.7222138 winter  moderate
81   81 13.6644541 winter  moderate
82   82 22.3453980 winter  moderate
83   83 18.4556571 winter  moderate
84   84  5.5702815 winter      good
85   85 23.7812263 winter  moderate
86   86  8.4406173 winter      good
```

```
87   87 10.7242514 winter      good
88   88  0.1539706 winter      good
89   89 16.2534373 winter  moderate
90   90  4.2754122 winter      good
91   91 13.6497579 spring  moderate
92   92 14.7529316 spring  moderate
93   93 23.8353172 spring  moderate
94   94 28.8464416 spring  moderate
95   95 23.3615585 spring  moderate
96   96 36.5483864 spring unhealthy
97   97 21.2852575 spring  moderate
98   98  6.5020059 spring      good
99   99  3.1902362 spring      good
100 100 23.8423710 spring  moderate
101 101 24.1509867 spring  moderate
102 102 13.3651090 spring  moderate
103 103 16.6996811 spring  moderate
104 104 36.0227203 spring unhealthy
105 105 23.9571838 spring  moderate
106 106 31.3495548 spring  moderate
107 107 17.1841359 spring  moderate
108 108 22.4460419 spring  moderate
109 109 21.3936013 spring  moderate
110 110 21.7237368 spring  moderate
111 111 18.5330577 spring  moderate
112 112 17.6426799 spring  moderate
113 113 30.1619830 spring  moderate
114 114 29.3316308 spring  moderate
115 115 25.9187114 spring  moderate
116 116 18.1765753 spring  moderate
117 117  0.0000000 spring      good
118 118  8.0783565 spring      good
119 119 27.1312134 spring  moderate
120 120 16.1646602 spring  moderate
121 121 18.5998714 spring  moderate
122 122 23.7478182 spring  moderate
123 123 11.7700436 spring      good
124 124 38.6144418 spring unhealthy
125 125 18.4778745 spring  moderate
126 126 14.1577523 spring  moderate
127 127 13.5029043 spring  moderate
128 128 28.1254933 spring  moderate
129 129 34.2868102 spring  moderate
130 130  9.1743739 spring      good
131 131 17.4290560 spring  moderate
```

```
132 132 14.7092843 spring  moderate
133 133 14.0314055 spring  moderate
134 134 18.6050713 spring  moderate
135 135 27.3276183 spring  moderate
136 136 23.7469129 spring  moderate
137 137  0.0000000 spring      good
138 138  4.2810278 spring      good
139 139  2.7119931 spring      good
140 140 30.4998249 spring  moderate
141 141 17.1097980 spring  moderate
142 142 11.8392452 spring      good
143 143  0.0000000 spring      good
144 144 15.1369313 spring  moderate
145 145  4.0626313 spring      good
146 146 28.2378825 spring  moderate
147 147 11.6673450 spring      good
148 148 21.1200278 spring  moderate
149 149 32.9801883 spring  moderate
150 150 33.5790305 spring  moderate
151 151 16.5977516 spring  moderate
152 152  3.4691087 spring      good
153 153 16.2170424 spring  moderate
154 154 28.4104183 spring  moderate
155 155 25.9105350 spring  moderate
156 156 16.5961788 spring  moderate
157 157 18.7056104 spring  moderate
158 158 14.7175867 spring  moderate
159 159 33.2173296 spring  moderate
160 160 17.1993077 spring  moderate
161 161  5.3585051 spring      good
162 162 23.6359386 spring  moderate
163 163 23.4271025 spring  moderate
164 164  9.2757360 spring      good
165 165 23.2001673 spring  moderate
166 166 19.1384421 spring  moderate
167 167 15.3056354 spring  moderate
168 168 16.3766101 spring  moderate
169 169 17.4437777 spring  moderate
170 170 19.8991517 spring  moderate
171 171  2.6050536 spring      good
172 172 18.2118540 spring  moderate
173 173 18.1784891 spring  moderate
174 174 34.7195203 spring  moderate
175 175 24.6967031 spring  moderate
176 176 34.3395998 spring  moderate
```

```
177 177 20.8215695 spring  moderate
178 178 17.3348558 spring  moderate
179 179 24.8093825 spring  moderate
180 180  6.4675217 spring      good
181 181 26.4358238 summer  moderate
182 182 14.4849586 summer  moderate
183 183 24.1697319 summer  moderate
184 184 18.9606674 summer  moderate
185 185 21.2242172 summer  moderate
186 186 12.5353087 summer  moderate
187 187  7.3884526 summer      good
188 188 20.1173387 summer  moderate
189 189 22.2701807 summer  moderate
190 190  6.0787455 summer      good
191 191 18.8579623 summer  moderate
192 192  3.8138633 summer      good
193 193  9.8127302 summer      good
194 194 17.3578746 summer  moderate
195 195 19.7778429 summer  moderate
196 196 20.0683817 summer  moderate
197 197 16.3063519 summer  moderate
198 198 34.5832885 summer  moderate
199 199 11.6719320 summer      good
200 200 13.5047934 summer  moderate
201 201 17.0019630 summer  moderate
202 202 20.5450058 summer  moderate
203 203 11.3588999 summer      good
204 204 14.1579020 summer  moderate
205 205 29.0591594 summer  moderate
206 206 20.1741675 summer  moderate
207 207 20.0849977 summer  moderate
208 208 10.9639398 summer      good
209 209 20.3017792 summer  moderate
210 210 10.0303680 summer      good
211 211  5.3880514 summer      good
212 212 18.7422367 summer  moderate
213 213  9.3588094 summer      good
214 214  0.8235893 summer      good
215 215 14.3200544 summer  moderate
216 216 14.9794991 summer  moderate
217 217 20.5329893 summer  moderate
218 218  7.1033463 summer      good
219 219 10.0199302 summer      good
220 220 23.6268966 summer  moderate
221 221 31.3736019 summer  moderate
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```
222 222 16.9013177 summer  moderate
223 223 17.9022961 summer  moderate
224 224 13.8300517 summer  moderate
225 225 18.8775911 summer  moderate
226 226 21.1976539 summer  moderate
227 227 17.8888145 summer  moderate
228 228 27.6266336 summer  moderate
229 229 20.6056680 summer  moderate
230 230 23.1527273 summer  moderate
231 231 29.1585422 summer  moderate
232 232 11.5147314 summer      good
233 233 32.7999945 summer  moderate
234 234 11.2443617 summer      good
235 235  5.3586265 summer      good
236 236 25.8959142 summer  moderate
237 237 27.6593702 summer  moderate
238 238 23.9812223 summer  moderate
239 239 22.4237188 summer  moderate
240 240 15.5960028 summer  moderate
241 241 26.4844109 summer  moderate
242 242 32.4939493 summer  moderate
243 243 19.2939626 summer  moderate
244 244 16.0027173 summer  moderate
245 245 27.7555677 summer  moderate
246 246 11.8366801 summer      good
247 247 33.9637606 summer  moderate
248 248 18.0161651 summer  moderate
249 249 20.4307085 summer  moderate
250 250  4.6151704 summer      good
251 251  4.8048202 summer      good
252 252 23.9287493 summer  moderate
253 253 38.9781355 summer unhealthy
254 254 25.3089870 summer  moderate
255 255 24.6805301 summer  moderate
256 256 11.5898601 summer      good
257 257 15.2773825 summer  moderate
258 258 34.9607239 summer  moderate
259 259 15.4479235 summer  moderate
260 260  8.0259603 summer      good
261 261  8.1747615 summer      good
262 262 19.2427918 summer  moderate
263 263 14.9431128 summer  moderate
264 264 14.0369127 summer  moderate
265 265 37.8878340 summer unhealthy
266 266 18.9660915 summer  moderate
```

```
267 267 23.9636020 summer  moderate
268 268 11.6173734 summer      good
269 269 31.5515879 summer  moderate
270 270 30.5764958 summer  moderate
271 271 18.1663632   fall  moderate
272 272 13.0582155   fall  moderate
273 273  0.0000000   fall      good
274 274 26.6022218   fall  moderate
275 275  1.2974309   fall      good
276 276 13.2707791   fall  moderate
277 277 10.2241505   fall      good
278 278 21.8557488   fall  moderate
279 279 15.4715656   fall  moderate
280 280  9.8973375   fall      good
281 281 19.2330767   fall  moderate
282 282 20.1748123   fall  moderate
283 283 15.2872611   fall  moderate
284 284 30.9175819   fall  moderate
285 285 17.0918663   fall  moderate
286 286 12.3866780   fall  moderate
287 287 24.3085859   fall  moderate
288 288 23.3285909   fall  moderate
289 289 15.6019094   fall  moderate
290 290 19.8032462   fall  moderate
291 291 17.2404892   fall  moderate
292 292 30.3761866   fall  moderate
293 293 11.8607867   fall      good
294 294 34.9131702   fall  moderate
295 295 13.9053250   fall  moderate
296 296 23.3967431   fall  moderate
297 297 10.8476440   fall      good
298 298  9.5501848   fall      good
299 299 28.1577918   fall  moderate
300 300 11.1819043   fall      good
301 301  8.8975061   fall      good
302 302 26.4954995   fall  moderate
303 303 18.9053705   fall  moderate
304 304 12.5618017   fall  moderate
305 305 15.0457516   fall  moderate
306 306 14.7191520   fall  moderate
307 307 32.2486524   fall  moderate
308 308 23.9568910   fall  moderate
309 309 13.9704307   fall  moderate
310 310 31.8146094   fall  moderate
311 311 23.2654003   fall  moderate
```

```
312 312 13.0529343   fall  moderate
313 313 19.4363175   fall  moderate
314 314 19.0775387   fall  moderate
315 315 25.0682258   fall  moderate
316 316 29.2153776   fall  moderate
317 317 21.0702522   fall  moderate
318 318 28.9155438   fall  moderate
319 319 15.9030225   fall  moderate
320 320 18.7254685   fall  moderate
321 321 14.8860059   fall  moderate
322 322 26.3134823   fall  moderate
323 323 24.0019548   fall  moderate
324 324 35.2775007   fall  moderate
325 325 29.8400112   fall  moderate
326 326 19.9938265   fall  moderate
327 327 21.2671537   fall  moderate
328 328  9.2484665   fall      good
329 329 17.4664192   fall  moderate
330 330 10.6304703   fall      good
331 331 30.3465334   fall  moderate
332 332 15.8669687   fall  moderate
333 333  2.6799686   fall      good
334 334 21.7583230   fall  moderate
335 335 13.6910148   fall  moderate
336 336 19.6882285   fall  moderate
337 337 22.0944514   fall  moderate
338 338 22.3138172   fall  moderate
339 339 11.7494570   fall      good
340 340 30.2207600   fall  moderate
341 341 27.3595575   fall  moderate
342 342 18.7731043   fall  moderate
343 343 35.8228563   fall unhealthy
344 344 26.4540714   fall  moderate
345 345  6.0731954   fall      good
346 346 21.1016823   fall  moderate
347 347 34.7147736   fall  moderate
348 348  8.3862148   fall      good
349 349 11.5906194   fall      good
350 350 22.2104096   fall  moderate
351 351  6.8779230   fall      good
352 352 31.0194722   fall  moderate
353 353 25.5517722   fall  moderate
354 354 21.8598353   fall  moderate
355 355 20.1863274   fall  moderate
356 356  6.6329519   fall      good
```

```
357 357 16.8808656   fall  moderate
358 358 12.8226946   fall  moderate
359 359 38.6946256   fall unhealthy
360 360  4.2516074   fall      good
361 361 10.5624495   fall      good
362 362 29.8393257   fall  moderate
363 363 35.7862656   fall unhealthy
364 364 17.2709180   fall  moderate
365 365 19.4143880   fall  moderate
```

Part 3.

```
#number of unhealthy days in each season

#initialize
winter_unhealthy <- 0
spring_unhealthy <- 0
summer_unhealthy <- 0
fall_unhealthy <- 0

for(i in 1:nrow(air_quality)) {
  if (air_quality$risk[i] == "unhealthy") {
    if (air_quality$season[i] == "winter") {
      winter_unhealthy <- winter_unhealthy + 1
    } else if (air_quality$season[i] == "spring") {
      spring_unhealthy <- spring_unhealthy + 1
    } else if (air_quality$season[i] == "summer") {
      summer_unhealthy <- summer_unhealthy + 1
    } else {
      fall_unhealthy <- fall_unhealthy <- 1
    }
  }
}

winter_unhealthy
```

```
[1] 1
```

```
spring_unhealthy
```

```
[1] 3
```

```
summer_unhealthy
```

```
[1] 2
```

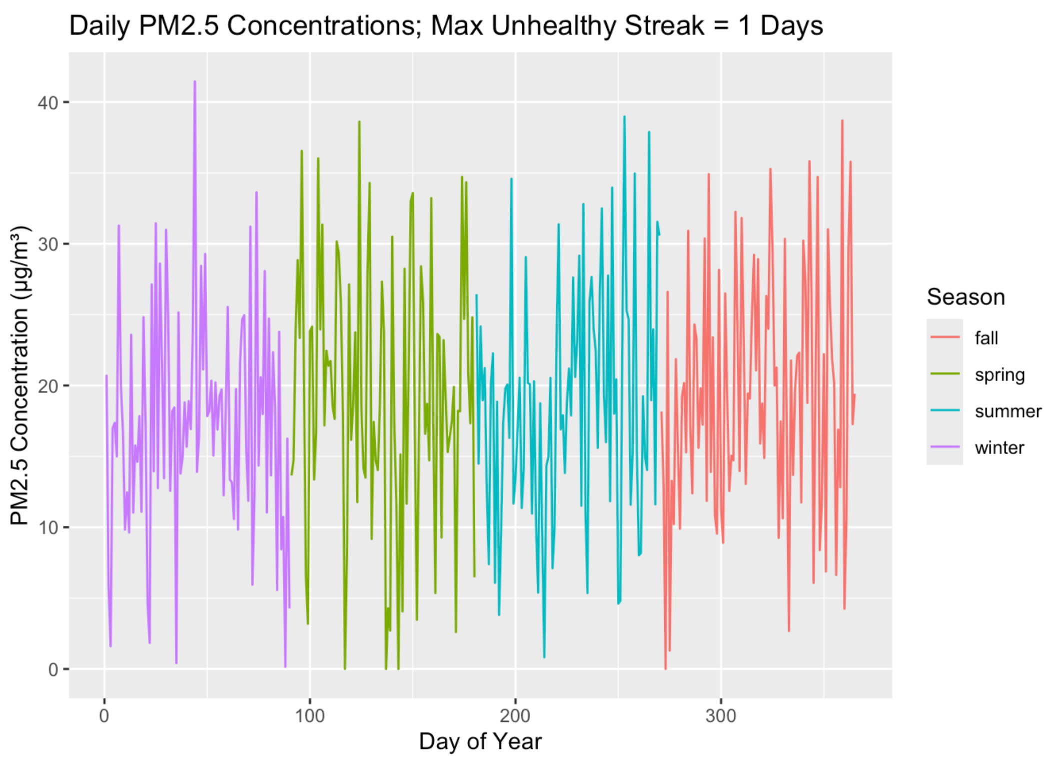
```
fall_unhealthy
```

```
[1] 1
```

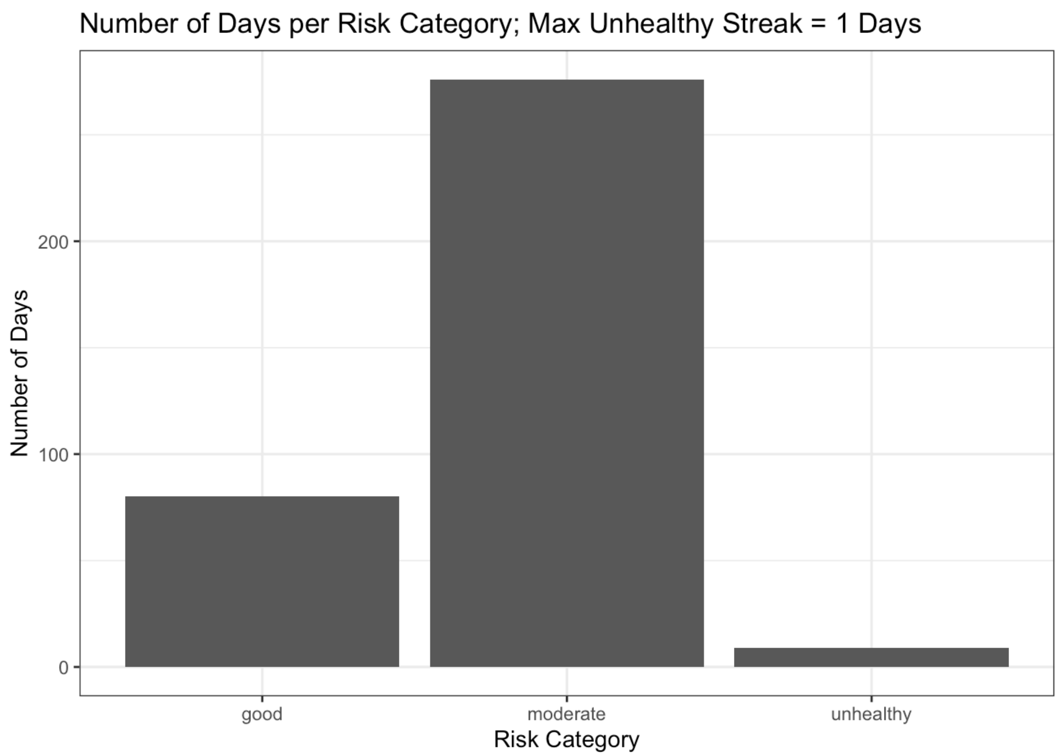
Part 4.

```
library(ggplot2)

ggplot(air_quality, aes(x = day, y = pm25, color = season)) +
  geom_line()+
  labs(title = sprintf("Daily PM2.5 Concentrations; Max Unhealtl
                         max_unhealthy_streak),
       x = "Day of Year",
       y = "PM2.5 Concentration (µg/m³)",
       color = "Season")
```



```
ggplot(air_quality, aes(x = risk)) +
  geom_bar() +
  labs(title = sprintf("Number of Days per Risk Category; Max Un
       x = "Risk Category",
       y = "Number of Days") +
  theme_bw()
```



The simulated PM2.5 concentrations remained mostly within the moderate air quality range throughout the year, although there were several small spikes that exceeded the unhealthy threshold of 35.4 µg/m³. Winter experienced the greatest number of unhealthy air quality days (4), followed by summer (3), spring (2), and fall (1). Summer also showed several elevated PM2.5 spikes, which may represent wildfire smoke during hot and dry months. Although the maximum consecutive unhealthy streak was only two days, short term exposure to high levels of particulate matter can negatively impact human health. Continued air monitoring is important for reducing exposure risks, especially during wildfire season.